

Constraint-Derived Geometry and Representation Systems for Biological and Artificial Intelligence Applications

1. FIELD OF THE DISCLOSURE

The present disclosure relates to computational systems, biological modeling, and artificial intelligence, and more specifically to systems in which **geometry and representation are derived from constraint-defined transformations rather than predefined metrics or embeddings**.

2. BACKGROUND

Existing computational and machine learning systems primarily rely on:

- metric-based embeddings
- similarity optimization
- distance-preserving representations

These approaches assume that:

geometry is learned from data.

However, biological systems operate under **explicit transformation constraints**, such as mutation rules in genetic systems, which define:

- which transitions are valid
- how transformations are structured

This suggests an alternative paradigm:

geometry is induced by constraints rather than learned from similarity

3. SUMMARY OF THE DISCLOSURE

The present disclosure provides systems and methods in which:

- a **constraint operator** defines permissible transformations
- a **geometry emerges from the structure of these transformations**
- a **representation is derived from the constraint structure**

In this framework:

- computation is defined by constraint-consistent transitions
 - geometry is not assumed or learned via distance
 - representation preserves transformation structure
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4. CONSTRAINT OPERATOR

In some embodiments, a constraint operator is defined as:

$$C(x) \rightarrow \{(x \rightarrow y, \tau)\}$$

Where:

- x is a state
- y is a reachable state
- τ is a transformation type

The constraint operator defines:

- valid transitions
 - transformation classes
 - asymmetries between transitions
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Key property

The constraint operator:

defines the structure of the system prior to any embedding or metric representation

5. CONSTRAINT-DERIVED GEOMETRY

A geometry is induced by the constraint operator across all states.

This geometry includes:

- adjacency (reachability)
 - transformation typing
 - directional asymmetry
 - higher-order structure
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Critical distinction

In conventional systems:

- geometry is defined by distance

In this disclosure:

geometry is defined by valid transformations

6. REPRESENTATION DEFINITION

A representation R is defined as:

$$R(x) = f(C(x))$$

Where the representation:

- preserves valid transitions
 - preserves transformation types
 - retains structural relationships
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Valid representation condition

A representation is valid if:

- valid transitions remain valid
- transformation classes remain distinguishable
- constraint structure is preserved

7. EMBEDDING VS CONSTRAINT REPRESENTATION

In some embodiments:

Embedding-based representations:

- derive structure from distances
- require optimization
- may not preserve transformation rules

Constraint-based representations:

- derive structure directly from transformations
 - require no optimization
 - explicitly encode valid transitions
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8. BIOLOGICAL EMBODIMENT — GENETIC CODE

In some embodiments:

- states correspond to codons
 - constraint operator corresponds to mutation rules
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Example constraint operator

$C(x)$ includes:

- all single-base mutations
 - transformation types:
 - position (P1, P2, P3)
 - mutation class (transition/transversion)
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Resulting structure

The constraint operator induces:

- synonymous mutation clustering
 - positional asymmetry
 - transformation-dependent effects
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9. FUNCTIONAL CONSEQUENCE MODELING

In some embodiments:

- functional outcomes depend on transformation types
 - constraint-defined transitions determine effects
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Key insight

Functional consequences are determined by how transformations act on states, not by similarity alone.

10. GEOMETRY–PHENOTYPE RELATIONSHIP

In some embodiments:

- distances are derived from paths through valid transitions
 - these distances correlate with functional differences
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Observed properties (Milestone 2)

- local regions exhibit near-flat structure
- boundaries are selectively pruned

- allowed transitions form consistent bridges
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Resulting geometry

piecewise flat geometry with selective connectivity

Key property

Constraint-defined geometry produces distances that more accurately reflect phenotype than unconstrained representations

11. COMPUTATION THROUGH CONSTRAINT GEOMETRY

In some embodiments:

- computation is defined as traversal of the constraint-derived geometry
 - valid outputs correspond to reachable states
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Critical statement

The constraint operator not only restricts transitions but defines permissible computation.

12. ARTIFICIAL INTELLIGENCE EMBODIMENT

In some embodiments:

- representations are generated from constraint structure
- learning occurs subject to constraint-defined transitions

Training mechanism

Learning may include:

- updating representations subject to constraint structure
- restricting optimization to valid transitions
- enforcing transformation consistency

Result

- improved stability
- improved generalization
- reduced invalid outputs

13. DISTRIBUTED AND MULTI-SYSTEM EMBODIMENTS

In some embodiments:

- multiple constraint-defined systems interact
- state spaces are composed or layered
- computation occurs across distributed systems

14. VARIATIONS AND EXTENSIONS

In various embodiments:

- constraints may be deterministic or probabilistic
 - constraints may be learned or predefined
 - state spaces may be discrete or continuous
 - transformations may be learned, stochastic, or composable
 - systems may integrate symbolic and subsymbolic representations
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15. EXPERIMENTAL VALIDATION (SUPPORTING EMBODIMENT)

In some embodiments:

- constraint-derived representations are evaluated against embedding-based systems
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Tasks include:

- mutation prediction
 - functional consequence classification
 - phenotype regression
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Observed outcomes:

- constraint-based representations improve functional prediction
 - transformation structure improves interpretability
 - geometry aligns with biological behavior
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16. GENERALIZATION AND SCOPE

The systems and methods described herein are not limited to biological systems.

The disclosed principles apply to:

- artificial intelligence
- biological modeling
- physical systems
- computational systems

Any system in which:

- relationships between states define valid transformations
- geometry is induced by those transformations
- computation is performed through traversal of such structure

falls within the scope of this disclosure.